

Deep-learning-based recognition of symbols and texts at an industrially applicable level from images of high-density piping and instrumentation diagrams

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ABSTRACT

Piping and instrumentation diagrams (P&IDs) are commonly used in the process industry as a transfer medium for the fundamental design of a plant and for detailed design, purchasing, procurement, construction, and commissioning decisions. The present study proposes a method for symbol and text recognition for P&ID images using deep-learning technology. Our proposed method consists of P&ID image pre-processing, symbol and text recognition, and the storage of the recognition results. We consider the recognition of symbols of different sizes and shape complexities in high-density P&ID images in a manner that is applicable to the process industry. We also standardize the training dataset structure and symbol taxonomy to optimize the developed deep neural network. A training dataset is created based on diagrams provided by a local Korean company. After training the model with this dataset, a recognition test produced relatively good results, with a precision and recall of 0.9718 and 0.9827 for symbols and 0.9386 and 0.9175 for text, respectively.

1. Introduction

Piping and instrumentation diagrams (P&ID) are regularly used to visualize process plant. A P&ID is created using a process flow diagram as a reference, and it describes the primary equipment and materials that constitute individual processes and provides details of the fluid flow. These characteristics mean that P&IDs are a fundamental transfer medium for plant design that provides the information necessary for detailed design, purchasing, procurement, construction, and commissioning decisions.

In a digital P&ID, the structured relationship between objects is expressed in a digital format, meaning that the information can be processed by computer. Digital P&IDs include symbols, connecting lines, outer borders, a title box, characters, and tables. The critical components of digital P&IDs are the symbols, which can be categorized into fitting, instrumentation, equipment, and off-page connector (OPC) symbols. Engineering, procurement, and construction (EPC) companies,

which are involved in plant design, procurement, and construction, generally work directly with digital P&IDs, and plant operators also tend to use digital P&IDs. However, for newly constructed plants, P&IDs are often created during the front-end engineering and design (FEED) phase or provided by equipment manufacturers in an image format that is subject to contractual agreements due to concerns over intellectual property protection. In addition, many P&IDs for older plants under long-term operation have been saved in image format and need to be digitized, while those operators seeking to improve or expand their plants may still be provided with P&IDs in image format by their external partners.

For these reasons, there is a dire need for the conversion of image-based P&IDs to digital P&IDs in the process industry. Digitization, which aims to produce a digital diagram that is identical to the original, can be achieved by extracting all of the necessary information from diagrammatic images using the recognition of high-level objects that are meaningful to the relevant application. The current manual conversion

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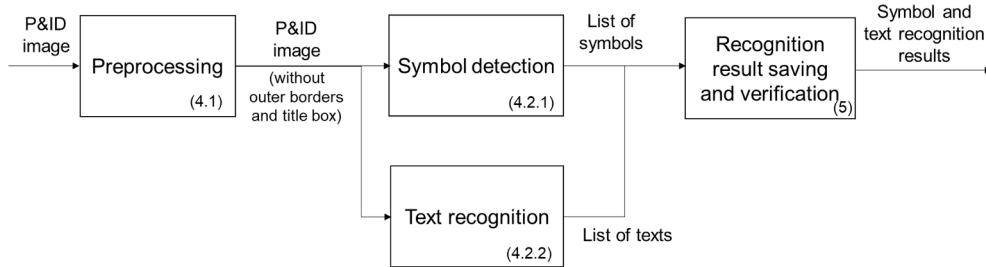


Fig. 1. Proposed method for symbol and text recognition from P&ID images using deep-learning technology.

process creates a digital P&ID via remodeling while referring to the original P&ID, but this process is time-consuming and error-prone, and the quality of the results depends on the skill level of the worker.

The present study proposes a novel method for symbol and text recognition using deep-learning technology. The input P&IDs are assumed to be a digital image generated using a P&ID authoring program that has no noise or transformation. The method consists of pre-processing the P&ID images, recognizing the symbols and text, and saving the results (Fig. 1). We focus particularly on the recognition of various symbols with different sizes and shape complexities that are present in high-density P&ID images to ensure that the proposed approach can be employed in the process industry. Additionally, we standardize the structure of the training dataset and the symbol taxonomy to optimize the functioning of the deep neural network.

The main contributions of this work are as follows:

- We present a P&ID recognition system that uses state-of-the-art deep-learning technology for symbol and text detection. To the best of our knowledge, it is the first industrially applicable P&ID recognition system that is robust for the use of high-density P&IDs.
- We introduce a small/large symbol network separation approach to enable size-variable symbol detection, adaptive non-max suppression (NMS) to improve the performance of industry-specific diagram configuration, and rotation estimation to robustly and accurately recognize rotated text in a diagram.
- We construct an industrial level P&ID training dataset composed of 75,031 symbols and 10,073 text instances by modeling digital P&IDs from P&ID images provided by a local Korean company.

The remainder of this paper is structured as follows. In Section 2, a review of previous literature on the extraction of information from P&ID images is presented, while Section 3 describes the structure of the training dataset used for the P&ID recognition process. In Section 4, we propose a method for symbol and text recognition in a P&ID. In Section 5, saving symbol and text recognition results followed by its verification and modification is presented, while Section 6 discusses experimental results for the use of the proposed method for the recognition of symbols and text in P&ID images in practical applications. In Section 7, a conclusion for our study is presented.

2. Related work

2.1. Extraction of design data from P&IDs

The essential elements of a P&ID are its symbols, the lines between these symbols, and the attributes assigned to these symbols and lines, such as text or signs (Tornbre, 1992). In a P&ID, the symbols and lines have particular meaning from an engineering and application perspective. Many studies have been published on the recognition of elements in various types of diagram, including electric, engineering, and logic diagrams and P&IDs (Benjamin et al., 1988; Fahn et al., 1988; Futatsumata et al., 1990; Kang et al., 2019; Kato & Inokuchi, 1990; Lee et al., 1990).

For the recognition of elements in a diagram, conventional image

processing techniques, including edge detection (Maini & Aggarwal, 2008), the Hough transform (Xu et al., 1990), and morphological operations (Wang et al., 1995), have been employed. Additionally, feature extraction from an image for the recognition of symbols and lines has been proposed (Ping Tian, 2013), with the extracted features compared to previously defined symbols and lines. However, traditional image recognition methods do not provide the robust recognition of symbols, lines, and text in image-based diagrams. The recognition success rate is reduced further if the individual objects interfere with each other, if the shape of an object is partially deformed, or if noise is present (e.g., stains).

With recent advances in deep-learning technology (Lecun et al., 2015), the use of convolutional neural network (CNN)-based deep-learning applications for the recognition of elements in diagrams has been reported (Fu & Kara, 2011; Rahul et al., 2019; Yu et al., 2019; Yun et al., 2020); however, the application targets have generally been limited to basic logic diagrams. In addition, even though the recognition target was a P&ID in Rahul et al. (2019), pre-processing of the diagrams and the recognition of lines and tables were not considered. A previous study by Yu et al. (2019) had proposed a recognition method for symbols, lines, and P&ID text, but this method was very time-consuming due to the use of a sliding window. Additionally, the text region is detected using a connectionist text proposal network (CTPN) (Tian et al., 2016), but the characters in the image could not be discerned. In Yun et al. (2020), the region-based CNN (R-CNN) suggested by Girshick et al. (2014) was used to recognize symbols in a P&ID, but the recognition of text and lines was not considered.

Previous approaches to P&ID recognition that apply deep-learning technology are limited and cannot be practically applied to the process industry. In particular, many of these approaches can only recognize around 10 symbols, while the symbols in a P&ID vary in size and shape complexity, and this has not been considered in previous studies. Additionally, the density of the elements in the P&IDs used in previous studies is lower than that used in the field. For this reason, the present study applies deep learning to high-density P&IDs containing symbols with different sizes and complex shapes to enable the recognition of symbols and texts that can be employed at the field level. This study also considers the structure of the training data and the standardization of symbol taxonomy to increase the applicability of P&ID recognition technologies in the field.

2.2. Object detection in images

Object detection technology is used to detect the location and categories of the objects in an image, which can be used in computer vision tasks to understand the image. Object detection pipelines based on traditional machine learning are designed to extract features such as the scale-invariant feature transform (Lowe, 2004), histograms of oriented gradients (Dalal & Triggs, 2005), and Haar-like features (Viola & Jones, 2001) from objects. Subsequently, classifiers such as support vector machine (Cortes & Vapnik, 1995), AdaBoost (Freund & Schapire, 1997), and random forest (Breiman, 2001) are trained to robustly classify the distribution of the features extracted from different objects. In the

application stage, the features extracted from an input image are passed through the classifier in order to detect a trained object.

Deep-learning-based object detection technology has been shown to perform much better than machine-learning-based technology in industrial applications (Zhang et al., 2020). In deep-learning-based technology, an artificial neural network is trained to learn how to extract the features from an image. Various CNN structures stacked with multiple convolution layers have been widely used to learn feature extraction from an image, including VGG (Simonyan & Zisserman, 2015), ResNet (He et al., 2016), MobileNet (Howard et al., 2017), GoogleNet (Szegedy et al., 2015), Xception (Chollet, 2017), and DenseNet (Huang et al., 2017). Research has also shown that classifying global features extracted from a CNN through a fully connected layer of stacked multi-layer perceptron (MLP) layers leads to high performance (Krizhevsky et al., 2012).

For object detection using a CNN, classification should be based on local rather than global features. Methods for deep-learning-based object detection can be categorized into the one-stage and multi-stage approaches. Multi-stage approaches that have been proposed include the faster R-CNN (Ren et al., 2015) and the cascade R-CNN (Cai & Vasconcelos, 2018), in which the region proposal network for the detection of a region of interest (RoI; where an object resides) and the classifier network for the classification of detected objects are kept separate. In contrast, the one-stage approach searches for an object's location and class without a region proposal network using the features obtained through a CNN, which retain spatial information. Various methods in one-stage approach have been proposed, such as YOLO (Redmon et al., 2016), SSD (Liu et al., 2016), RetinaNet (Lin et al., 2017), feature selective anchor-free (FSAF) method (Zhu et al., 2019), and generalized focal loss (GFL) (Li et al., 2020). Multi-stage approaches tend to be more precise, although they demand more operations than a one-stage approach. Recently, however, one-stage approaches have exhibited performances that are nearly as good as those of multi-stage approaches.

Previous deep-learning-based object detection research has addressed many of the challenges associated with object detection. RetinaNet proposed a focal loss function to resolve the class imbalance problem that is associated with the poor performance of the one-stage approach, and experiments showed an improvement in this regard. Focal loss is an appropriate approach for data with a severe class imbalance. FSAF addressed a limitation of the anchor-based methodology, which heuristic selection of feature is required during the learning stages, and proposed a methodology that automatically selects appropriate features using anchor-free branches. The performance of the one-stage approach differs greatly depending on the anchor selection method when objects have a wide size distribution, so an anchor-free approach was designed to offer a more robust performance. GFL extends the focal loss function in RetinaNet by using a distribution function for object classification and object location regression.

In this study, we investigated the application of deep-learning-based object detection for the recognition of symbols in a P&ID. In previous papers, commonly used image datasets were used to assess the performance of deep-learning-based object detection technologies, but there is no guarantee that proposed methodologies will be equally useful for P&ID images. This is because deep-learning technology is greatly affected by the characteristics of the training data and its distribution, so its performance may vary depending on the dataset. This study thus investigated symbol detection in P&IDs, focusing on a one-stage approach under various conditions, and proposed strategies to optimize its performance. Two networks were configured to compensate for the performance degradation caused by differences in the symbol sizes in P&IDs. Post-processing and augmentation of the data were also used to improve the recognition performance for high-density symbols.

2.3. Text recognition in images

Text recognition within images based on optical character recognition (OCR) technology has been investigated for use in areas including machine translation and image searches. Similar to object recognition, technologies for recognizing the characters in an image pass the features extracted from the image to a classifier trained to recognize individual characters. However, in text recognition, the text, as a collection of characters with meaning, is the target of the recognition process, not individual characters. To recognize a text string, the characters that make up the text can be clustered, meaning each character in the text should be separable for recognition. There are a number of commercial software packages that are available for this, such as FineReader (ABBYY, 2021) and OmniPage (Kofax, 2021), as well as open-source software such as GOCR (Schulenburg, 2018).

Text recognition consists of two steps: text region detection, in which the image is searched to identify areas with consecutive characters, and the recognition of the characters in the identified areas. In the first step, connected-component analysis or the sliding window technique has been traditionally used. Recently, methods combining a CNN and a recurrent neural network (RNN) have been proposed to improve text region detection and text recognition within images (Chung et al., 2014; Graves et al., 2013; Huang et al., 2015). An RNN is suitable for the processing of ordered data, such as voice and text, because the identification of the current data point is dependent on the previous and following data points.

This study employed easyOCR (Jaided AI, 2021), which offers an integrated framework of text region detection using CRAFT (Baek et al., 2019) and text recognition using CRNN (Chi et al., 2015). Tesseract (Smith, 2007) was also used for text recognition to improve the text recognition performance for P&IDs. We avoided commercial software because we could not freely modify or train the underlying algorithms, while a comparative analysis of GOCR and Tesseract found that Tesseract offers a superior performance (Dhiman & Singh, 2013). Text recognition for P&IDs is less complex than text recognition in natural images because a P&ID is a formatted document that does not have a complex background. We used easyOCR primarily for text region detection and Tesseract for character recognition. The recognition results were analyzed and evaluated using state-of-the-art text recognition performance measures.

3. Training data for recognition of P&IDs

3.1. Structure of the training data

To ensure the high recognition accuracy of deep-neural-network-based image classification, the training data should be of good quality. Since the public release of the ImageNet in 2009 (Deng et al., 2009), larger datasets with various types of label information have been released, including OpenImage (Benenson et al., 2019), Microsoft COCO (Lin et al., 2014), and LVIS (Gupta et al., 2019). In this study, however, the construction of additional training data was necessary for the recognition of symbols, characters, and lines in a P&ID, so we defined the data structure to identify the information required for object recognition and trained a model based on this.

A P&ID consists of various image components such as symbols, texts, outer borders and a title. In this study, the objects to be recognized within a P&ID are symbols, texts, and lines. The preliminary information to be extracted for the recognition of symbols and text objects was the class (i.e., label), location, and orientation information. The class information was taken to be the name of an object following P&ID standards, while the location information for the object was the top-left and bottom-right coordinates within the object's bounding box. The orientation information for symbols and text was represented by the four main directions (i.e., north, east, south, and west), with north as a reference, and their shapes followed P&ID standards. The information

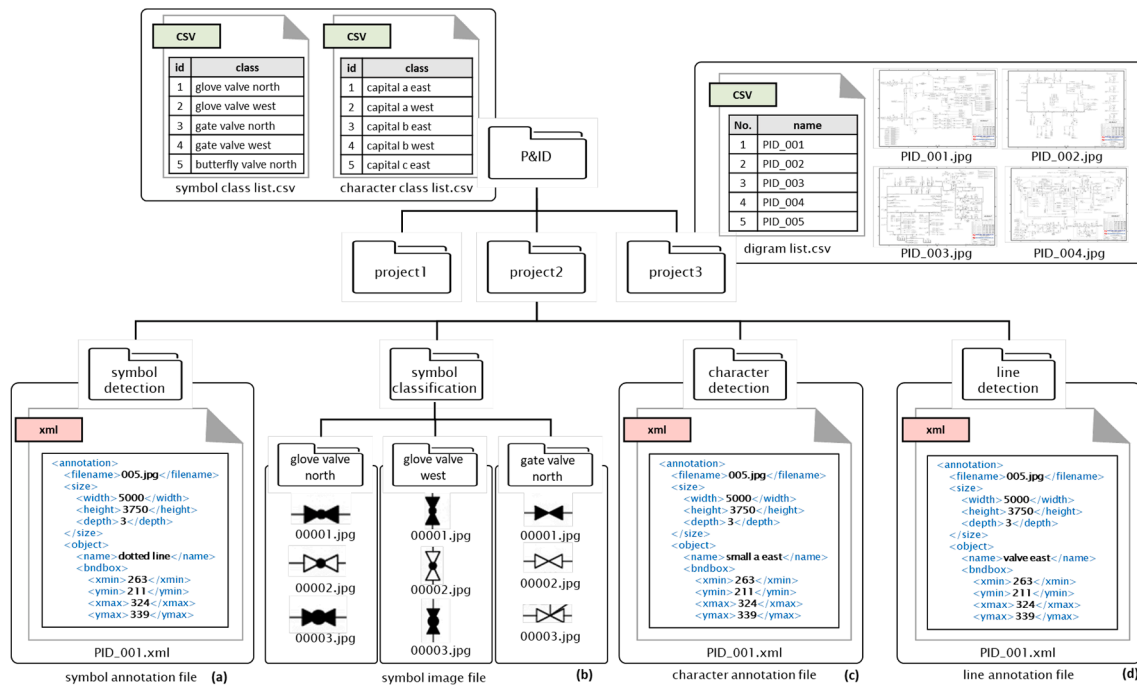


Fig. 2. Composition of the training data for object recognition in P&IDs.

Table 1
Scope of P&ID symbol standards.

	PIP PIC001	ANSI/ISA 5.1	ISO 15519-2	ISO 10628-2
Scope	P&ID (process industry)	Instrumentation	P&ID (process industry)	Equipment and piping (chemical and petrochemical Industries)
Equipment	9 categories, 48 items	–	1 category, 6 items	20 categories, 198 items
Piping	6 categories, 63 items	–	2 categories, 41 items	5 categories, 60 items
Instrumentation	7 categories, 92 items	11 categories, 176 items	3 categories, 52 items	–
Other	1 category, 6 items	–	–	4 categories, 38 items
Total	209 items	176 items	99 items	296 items

for line extraction included the class, the beginning and end points, the point shape, and the position and direction of the flow symbol. The shape of the line's end point indicates the presence or absence of an arrow. The flow symbol designates the direction of the fluid flow within a pipe, which was also represented by the four main directions.

The storage structure of training data from P&IDs for the deep neural network are presented in Fig. 2. At the top level is a training data folder structure, with the P&ID folder containing a file with a list of classes of symbols and texts. A symbol's class is defined by its type and direction, and the text class is defined according to the presence of lowercase or uppercase characters and the direction of the text. Diagrams created for the project and a file containing the list of diagrams can be found in the project folder. Training data are stored in the project folder because P&ID standards can differ between projects. The subfolders within the project folder include those for symbol detection (Fig. 2a) and the storage of training data for symbol classification (Fig. 2b), text recognition (Fig. 2c), and line recognition (Fig. 2d). In the training data folder

for each object, an annotation file is provided for each diagram. The annotation file structure is defined by referring to the data structure of the Visual Object Classes (VOC) challenge (Everingham et al., 2010), an object recognition competition hosted by Pattern Analysis, Statistical Modeling, and Computational Learning (PASCAL). Symbol image files extracted from each diagram are also stored in the training data storage folder.

3.2. Standard-based selection of labels for P&ID symbols

Before building the dataset, it was important to establish a P&ID symbol taxonomy using the correct terms and classifications for plant engineering in order to facilitate user understanding of P&ID recognition technology. In the study, the P&ID symbol taxonomy was defined based on well-known P&ID symbol standards.

There are several standards available for the shapes of P&ID symbols. The four that are most widely used in commercial P&ID authoring tools are PIP PIC001 Piping and Instrumentation Diagram Documentation Criteria (Process Industry Practices, 2008), ANSI/ISA 5.1 Instrumentation Symbols and Identification (ISA, 2009), ISO 15519 Specifications for Diagrams for Process Industry-Part 2: Measurement and Control (ISO, 2015), and ISO 10628 Diagrams for the Chemical and Petrochemical Industry – Part 2: Graphical Symbols (ISO, 2012). PIP PIC001 is the standard used in the process industry, including chemical and pharmaceutical engineering. PIP PIC001 consists of equipment symbols (9 categories, 48 items), piping symbols (6 categories, 63 items), instrumentation symbols (7 categories, 92 items), and other symbols (6 items). ANSI/ISA-5.1 is a widely used standard in all the fields related to instrumentation and control systems. It only consists of instrumentation symbols (11 categories, 176 items). As with PIP PIC001, ISO 15519-2 and ISO 10628-2 are well-known standards in the process industry and consist of equipment, piping, and instrumentation symbols. A summary of each standard is provided in Table 1.

Each standard was reviewed, focusing on details of the symbol classification, to select the reference standards for defining the P&ID symbol taxonomy. Furthermore, based on brief interviews with plant engineers, the connection between the symbols used in the standards and real-life cases was examined.

Table 2
Number of symbol labels defined.

Symbol type	Equipment	Piping	Instruments	Total
Number of standard-referenced symbol labels	198	63	77	338
Number of additional symbol labels missing in standards		13		13
Number of additional symbol labels from collected dataset	22	11	2	35
Total number of symbol labels defined	220	87	79	386

ANSI/ISA 5.1 does not provide information about equipment symbols, while ISO 10628-2 provides the most comprehensive equipment symbol classifications and its symbols are most like the P&ID symbols used in the field. ANSI/ISA 5.1 does not provide piping symbols, while PIP PIC001 and ISO 10628-2 provide similar levels of piping symbol classification. ISO 10628-2 does not provide instrumentation symbols, and ANSI/ISA 5.1 provides the most detailed instrumentation symbol classification. PIP PIC001 provides the second most detailed instrumentation symbol classification. However, ANSI/ISA 5.1 recognizes the individual parts of an instrumentation symbol as independent symbols. Thus, in terms of the instrumentation symbols used in P&IDs, ANSI/ISA 5.1 and PIP PIC 001 provide the same level of symbol classification.

In conclusion, ISO 10628-2 is suitable for equipment symbol classification, both PIP PIC001 and ISO 10628-2 are suitable for piping symbol classification, and PIP PIC001 and ANSI/ISA 5.1 are suitable for instrumentation symbol classification. Therefore, in the study, PIP PIC001 and ISO 10628-2 were selected to completely cover the equipment, piping, and instrumentation symbol classifications. ISO 10628-2 was used as a reference for equipment symbol classification, and PIP PIC001 was used for piping and instrumentation symbol classification. As a result, a P&ID symbol taxonomy consisting of 338P&ID symbols was first defined based on the standards.

However, two sets of symbol labels were further added in this study. Firstly, nozzles that are often used in the industry were not defined in the referenced standards. Therefore, we defined 13 nozzle symbol labels and added them to our symbol labels. Secondly, while collecting the P&ID image dataset, symbols that were not included in our standard-based symbol classification were discovered. This was because customized

symbols were used to create P&IDs depending on constructed projects. Based on the collected P&ID dataset, 35P&ID symbols were additionally defined so that recognition experiments could be performed even for customized symbols. Thus, a total of 386 symbol labels were defined in this research. Table 2 shows number of symbol labels defined in our symbol taxonomy. We provided a table of entire symbol labels in the [supplementary material](#).

4. Symbol and text recognition in P&ID images

To recognize symbols and text, state-of-the-art deep-learning technologies were used. This section is structured as follows. In Section 4.1, pre-processing of the P&ID image for the outer border and title box extraction is introduced. In Section 4.2, the proposed symbol detection method is introduced and compared with previous deep-learning-based object detection methods, and small/large symbol network separation and an adaptive NMS approach are proposed. In Section 4.3, the proposed text recognition method is introduced and a rotation estimation pipeline is presented.

4.1. Image pre-processing

The outer borders and title box of a P&ID are not significantly different from symbol, text, and line objects, but they may lower recognition accuracy, thus removing them before the recognition process may be required. The shape of the outer border and title box of P&IDs may vary between projects. The outer border typically consists of single or double solid lines with minor guidelines in some cases. The title box can be located in the bottom left, center, or right. A comment area is usually found on the right side of the diagram in the title area.

Fig. 3 presents the identification and removal process for the outer borders and title box of a P&ID. First, the bottom 1/5th of the diagram is cut off from the rest of the diagram to reduce the computation burden for the image processing process. All of the horizontal lines in this isolated region are then extracted, and the longest of these lines is extracted (Fig. 3b). Following this, the single point is randomly chosen from the identified longest line. Because the title box is connected to the outer borders, it can be separated by extracting all black pixels connected to the points that make up the identified outer border (Fig. 3c). Morphology operations (e.g., erosion and dilation) are then applied to

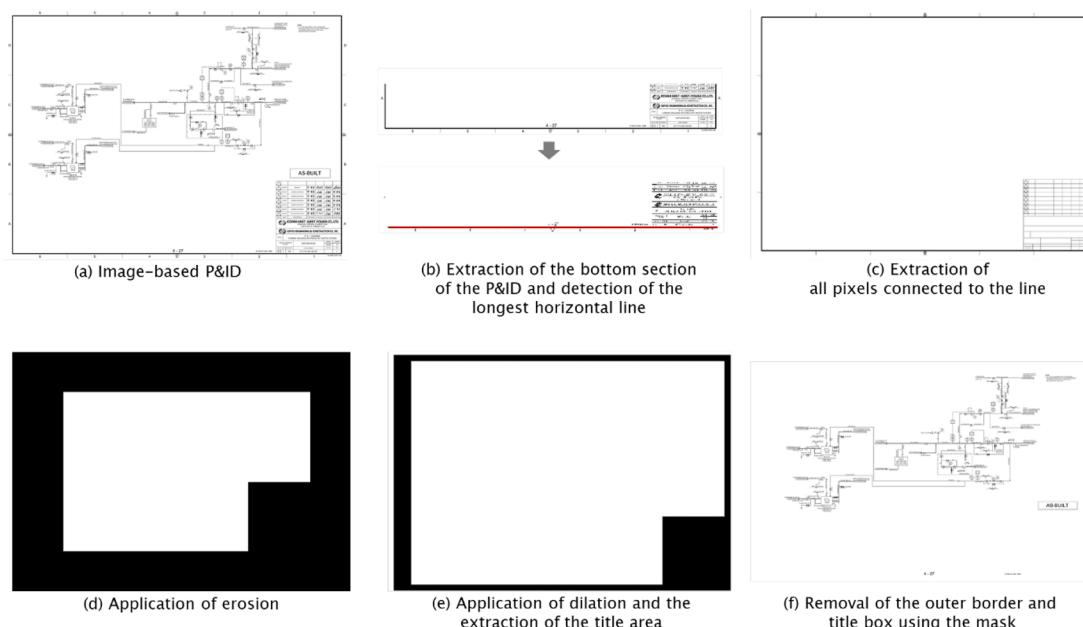


Fig. 3. Removal of the outer border and title area in a P&ID.

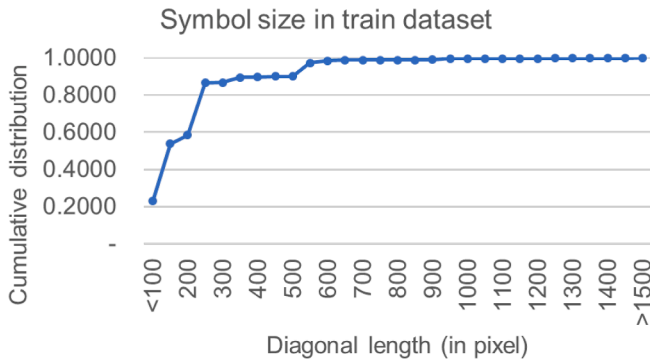


Fig. 4. Distribution of the symbol sizes in the training dataset.

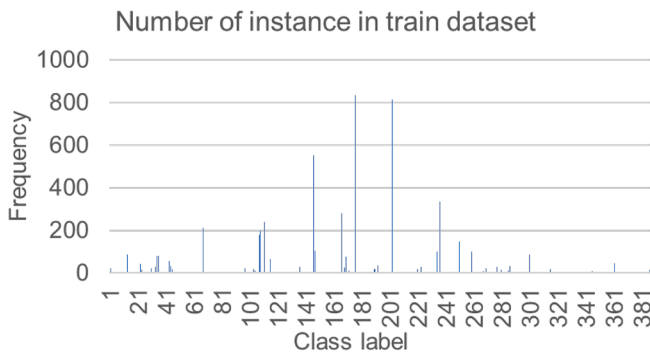


Fig. 5. Number of class instances in the training dataset.

the extracted black pixels to identify the outer border and title area. Finally, the identified area is removed from the P&ID.

A morphology operation is an image-processing technique widely used to transform the shape of an object in an image during pre- or post-processing. The most basic morphological operations are erosion and dilation. Erosion replaces the current pixel values with the minimum of all pixel values in the kernel region; as seen in Fig. 3(d), the size of the black region is increased, while that of the white region is reduced. In contrast, dilation replaces the current pixel values with the maximum of all pixel values in the kernel region; thus, the size of the white region is increased, while that of the black region is reduced in Fig. 3(e). This black area represents the outer borders and the title. Performing dilation after erosion is referred to as performing open operations.

4.2. Symbol detection

4.2.1. Deep-learning-based object detection

The P&ID symbol dataset constructed in this study (see Section 6.1) differs from existing image datasets such as COCO in that the P&ID symbol data consist of relatively simple images with noiseless white backgrounds while also experiencing a class imbalance problem (i.e., substantial deviations in the training data depending on diagrammatic

characteristics from class to class). In contrast, instances are distributed more evenly for all classes in existing datasets. In addition, the size of the objects varies dramatically in P&IDs. A simple image with a noiseless white background reduces the chances of false positives and eliminates the need for hard negative mining (Siva et al., 2012). However, the class imbalance and wide distribution of object sizes complicate object detection.

Fig. 4 presents the distribution of object sizes for the P&ID symbols in the training dataset. The most common diagonal length of the symbols is between 100 and 150 pixels, while 90% of the symbols have a size of 450 pixels or less. Fig. 5 presents the number of instances for each class, with the most common class having 500 to 800 instances. The training dataset consisted of 73 symbol labels out of 386 symbols. It should be noted that 76 symbol labels existed in the entire dataset, while 3 symbol labels only existed in the test diagram that was excluded in the training process. Fig. 6 displays the top three classes with the largest number of instances: the flange, globe valve, and gate valve. The relative size of the symbols is the same as the size in the actual diagram.

A preliminary experiment was conducted on three state-of-the-art methodologies that address class imbalance and size variation issues: RetinaNet, FSAF, and GFL. The objective of the preliminary experimentation was to identify the most effective object detection method when various backbone networks and resolutions for the input P&ID image are provided. As pointed out in Section 2, RetinaNet, FSAF, and GFL offer specific advantages for symbol detection with P&IDs. Thus, quantitative analysis of symbol detection performance is required before the symbol detection process can be designed in detail. In the P&ID symbol dataset, many of the symbols frequently lie within a relatively narrow area, while the box coordinates assigned to each symbol in the dataset may be ambiguous. Thus, it was assumed that GFL would be able to produce more robust results.

9 out of the 82 diagrams in the training dataset were used as a test set to estimate performance, with 90% of the remaining diagrams used as training data and 10% used as validation data. The performance for each learning epoch was measured based on the validation dataset. We trained each experimental case for 20 epochs using an Adam optimizer with a learning rate of 0.00035. A detailed explanation of the training dataset is introduced in Section 6.1. Training and prediction were conducted by dividing the diagrams into 800×800 sections because the memory resources were unable to handle the entire diagram at once. This process led to poor performance when detecting large symbols in the diagrams. This problem is addressed further in Section 4.2.2.

The first experiments were conducted to quantitatively compare the

Table 3

Performance of object recognition methods for P&ID symbol data.

Method	Resolution	Precision	Recall	AP	AP ₅₀	AP ₇₅
RetinaNet	0.5	0.9302	0.9777	0.6290	0.8760	0.7290
	1.0	0.9487	0.9862	0.7040	0.8490	0.8030
FSAF	0.5	0.9500	0.9770	0.6540	0.9030	0.8010
	1.0	0.9788	0.9804	0.7550	0.8660	0.8350
GFL	0.5	0.9376	0.9836	0.6480	0.8950	0.7780
	1.0	0.9701	0.9899	0.7520	0.8990	0.8540

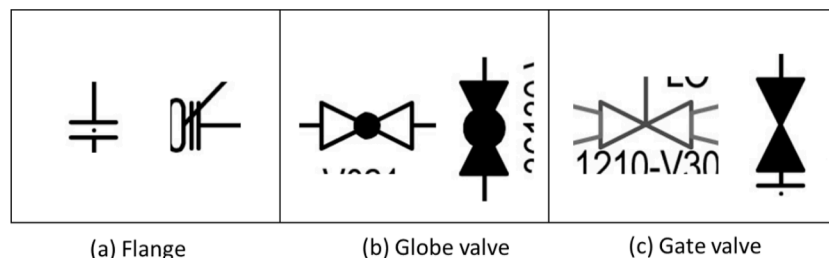


Fig. 6. Representative images of three most frequent symbols: (a) flange, (b) globe valve, and (c) gate valve.

Table 4

Performance of GFL using different backbone networks for P&ID symbol data.

Method	Backbone	Precision	Recall	AP	AP ₅₀	AP ₇₅
GFL	ResNet18	0.8880	0.9440	0.7080	0.8560	0.7960
	ResNet50	0.9701	0.9899	0.7520	0.8990	0.8540
	ResNet101	0.9535	0.9880	0.7390	0.8870	0.8500

performance of RetinaNet, FSAF, and GFL for the original diagram resolution (scale 1.0) and for the diagrams with half the length and width (scale 0.5). These resolutions were used to observe the effect of the information loss on the performance of these approaches. If a lower resolution does not affect performance, low-resolution diagrams can be used to increase the prediction time. As the backbone network, ResNet50 was used for all three methods in the initial experiment.

The results of the experiment are presented in Table 3. The item with the highest performance in each column is bolded and underlined, while the second-highest is bolded and italicized. According to Table 3, GFL had a higher overall performance for a resolution of 1.0 and had the highest performance under the stricter intersection over union (IOU) setting (AP₇₅).

We then examined GFL performance for a resolution of 1.0 with a backbone network of ResNet18, ResNet50, and ResNet101. ResNet101 generally produces a higher performance than ResNet50 and ResNet18 it has deeper configured layers. However, this difference in performance is observable only if a sufficient amount of training data is used for the learning process. Thus, experiments were conducted to directly verify the performance of the P&ID symbol data generated in this study.

Table 4 presents a summary of the experimental results, with ResNet50 showing the highest performance. Overall, the two preliminary experiments confirmed that using a resolution of 1.0 produced the highest performance for a network configured based on GFL with ResNet50 as the backbone. All of the symbol detection methods employed in subsequent analyses were thus based on these conditions.

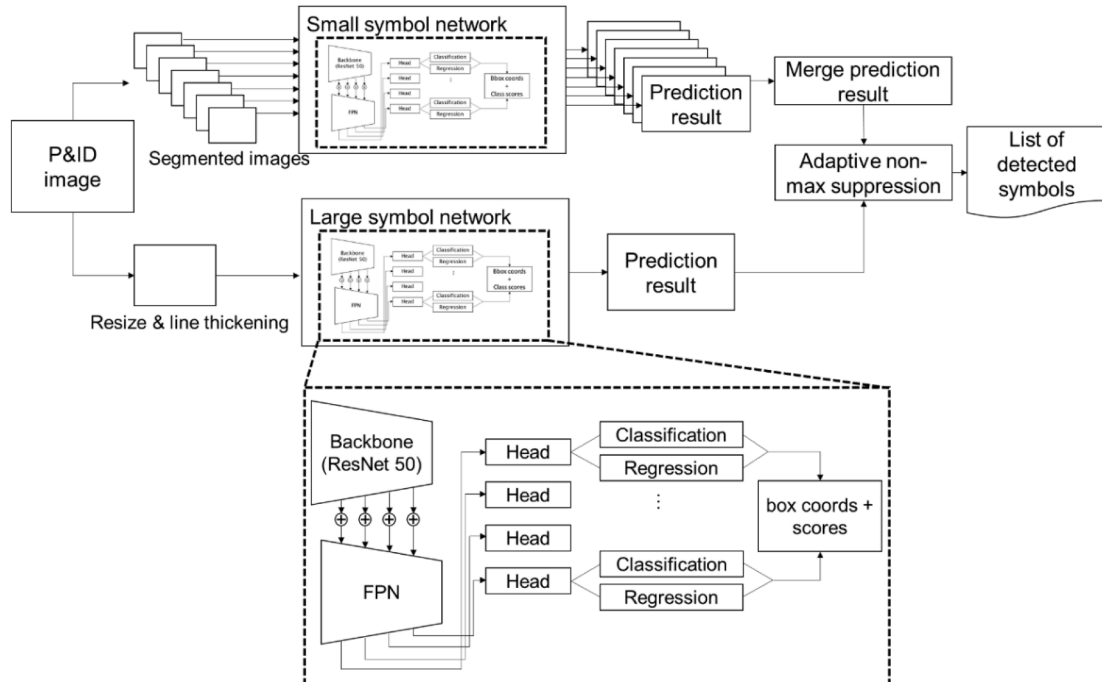
4.2.2. Proposed symbol detection process

When applying the GFL methodology to actual P&ID symbol detection, additional pre-processing and post-processing are required. For the

P&ID symbol detection process used in this study, 1) large- and small-symbol detection networks were implemented separately and their results merged, and 2) an adaptive NMS process was employed to improve the performance during the merging process. A schematic of the overall symbol detection approach, including the two proposed processes, is presented in Fig. 7. Each input diagram was subject to two separate operations: it was divided into several sub-diagrams for small-symbol network learning and prediction and resized for large-symbol network learning and prediction. The same network structure proposed for the GFL method was used for both the small- and large-symbol networks (Fig. 7), but learning was conducted separately. During the prediction process, the diagram was divided and resized using the same pre-processing methods used for learning. The small-symbol network results were subject to additional merging, converting the coordinates of the divided sub-diagrams into those of the original diagram. The final result was derived from the proposed adaptive NMS.

4.2.2.1. Small/large symbol network separation. As in our preliminary experiment, the entire diagram could not be fed into the network at the learning or prediction stage due to limited memory resources. This is a practical issue for the industrial application of recognition systems; it is necessary to keep the system requirements for the proposed P&ID recognition system as low as possible so that it can be used in a PC environment, thus increasing its usability. In this study, the images used for P&ID symbol recognition had a resolution of 9933×7016 with a DPI of 300. Previous studies would have downscaled the size of these images for object detection, but for the method proposed in the present paper, downscaling would have reduced performance because of the information lost in the downscaling process. For example, learning the features of a symbol consisting primarily of thin lines would be challenging due to the loss of pixel information. Thus, the P&ID image was divided into multiple smaller sections with some overlapping regions, with learning and prediction conducted on each of these smaller sections.

For the learning and prediction of the divided diagrams, large symbols with a diagonal length of > 700 pixels were excluded from the learning process. If the box for an object crossed over the border of the divided section, it was also excluded from learning because otherwise it was difficult to correctly detect the symbol box, and inaccurately

**Fig. 7.** Proposed P&ID symbol detection process.

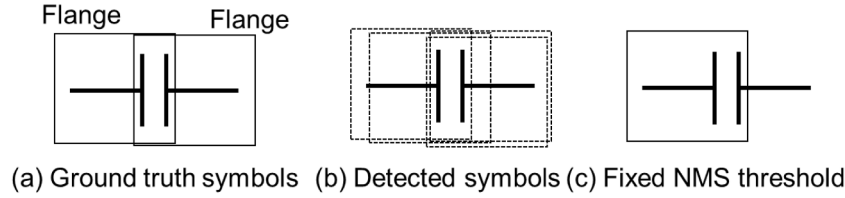


Fig. 8. Problem with the use of a fixed NMS threshold.

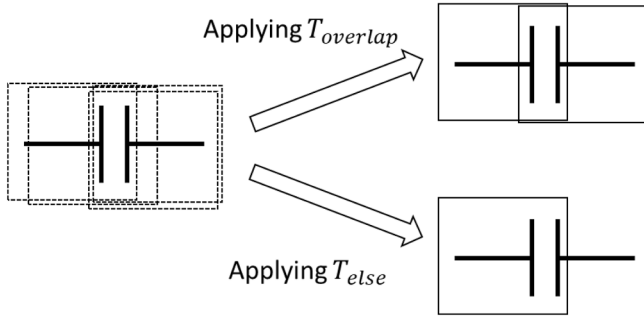


Fig. 9. Effect of applying a high threshold for self-overlapping symbols.

Table 5
Representative results for Tesseract and easyOCR character recognition.

Text Images	Tesseract Recognition	easyOCR Recognition
	S-199-MV-0003	\$-199-1 /-0003
	S-150-708-FH344-A1DT0-NA-57	\$-150-708-F 344-A1Dt0-//4-57
	S-30-199-PL330-A6DJ0-EI	\$-80-199-P.330-A6DJ0-ET

detected boxes decreased the performance during the merging process. Therefore, we proposed the use of a separate network that can process large symbols for learning and prediction. For this large-symbol network, learning was performed only on filtered symbol data with a diagonal length of >700 pixels in the original diagram. The image input was created by vertically and horizontally reducing the diagram to a 0.125 scale to learn the diagram without splitting, during which the erosion operation was used to thicken the lines to guarantee that no information was lost. Similarly, for the prediction stage, identical downscaling and line thickening operations were applied to each image and used as input to the large-symbol network.

4.2.2.2. Adaptive NMS. The prediction output for the symbol detection network is the box coordinates of the image area determined to contain symbols, the class of the symbol in the corresponding image area, and the confidence score for the prediction. To extract only the high scores for multiple box coordinates derived from a single object instance, a greedy NMS algorithm is generally used. The greedy NMS algorithm calculates the IOU between those boxes detected as an identical object to decide on the box with the highest score from among those boxes exceeding a specific IOU threshold; the rest of the boxes are deleted.

The performance of P&ID symbol detection is lower for the greedy NMS because of limitations associated with the use of a single IOU threshold value; a high threshold may not remove all duplicate boxes, while a low threshold may remove some correctly detected boxes. In Fig. 8, a low threshold results in the removal of a correctly detected symbol in the case of self-overlap, where two or more symbols in the same classification overlap, which is the case for certain symbols in a P&ID, including flanges and caps. In the NMS process (Fig. 8c), one of the two correct symbols that are present in Fig. 8(a) are removed.

In the adaptive NMS process, symbols in a P&ID with self-overlap are discerned using a heuristic approach. In our experimentation, we first compiled a list of self-overlapping symbols by calculating the IOU between the ground-truth bounding boxes for all symbol instances in the training dataset. Note that the validation dataset was not included in this process to fairly calculate the effect of the adaptive NMS approach. The list of symbol classes that had an IOU of >0 was defined as set $\mathcal{C}$ and used in the experiment. The detection of self-overlapping symbols was removed during NMS using Eq. (1):

$$\text{Remove detection if } \begin{cases} IoU(b_i, b_j) > T_{overlap}, & C(b_i) = C(b_j) \text{ and } C(b_i) \in \mathcal{C} \\ IoU(b_i, b_j) > T_{else}, & C(b_i) = C(b_j) \text{ and } C(b_i) \notin \mathcal{C} \end{cases} \quad (1)$$

where $b_i = \{x_{min}^i, x_{max}^i, y_{min}^i, y_{max}^i, class^i\}$ contains the box coordinates and the score information of the detected symbol, $IoU(b_i, b_j)$ is a function of the two detection results b_i and b_j calculates IOU, and $C(b_i)$ is a function that returns $class^i$. Our experiment confirmed the best performance with 0.0 for T_{else} and 0.2–0.5 for $T_{overlap}$. As a result, all of the symbols with self-overlap were detected (Fig. 9). If two detections were from different classes (i.e., $C(b_i) \neq C(b_j)$), they were not removed by NMS.

4.3. Text recognition

Deep-learning-based text recognition technology was applied to detect the location of text and to recognize the characters in text from an image. In this study, easyOCR was used for text region detection, and Tesseract was used to recognize the characters within that regions. The easyOCR framework internally conducts text region detection and character recognition using CRAFT and CRNN methods and then employs post-processing. In this study, only the results for the text region derived by the easyOCR framework were used; Tesseract was used for character recognition separately. Our preliminary experiment confirmed that Tesseract had a higher character recognition performance for P&ID images than did easyOCR. Specifically, easyOCR often failed to recognize characters such as 'S', 'M', 'H'. A comparison of representative character recognition results for Tesseract and easyOCR is presented in Table 5. Character recognition using Tesseract after conducting CRAFT-only text region detection was also considered. However, experiments confirmed that text region merging during post-processing with the easyOCR framework improved recognition performance.

The overall text recognition process is summarized in Fig. 10. A P&ID is a noiseless digital document that produces promising results with the easyOCR framework and Tesseract when used with weights learned from images of English text. The learned weights provided by Tesseract exhibited slightly better performance for P&ID images compared to the weights learned using the P&ID text dataset that was constructed in this study (see Section 6.1). The weights from Tesseract had about 1% and 2% higher precision and recall, respectively. It can thus be concluded that a large amount of text training data is effective even if the training data is not from P&IDs. As with symbol detection, the diagram was segmented first, and then the text region and recognition scores were determined using easyOCR in each segmented image. After that, these

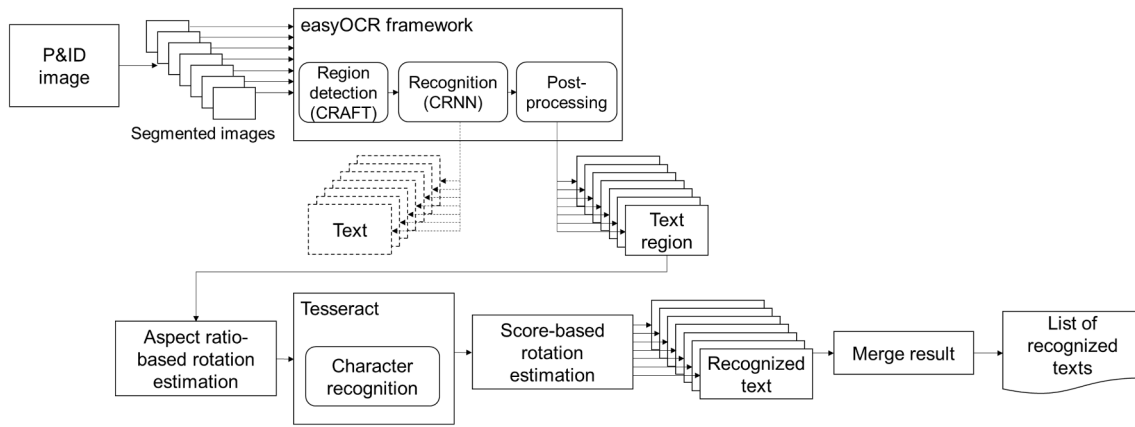


Fig. 10. Proposed P&ID text recognition process.

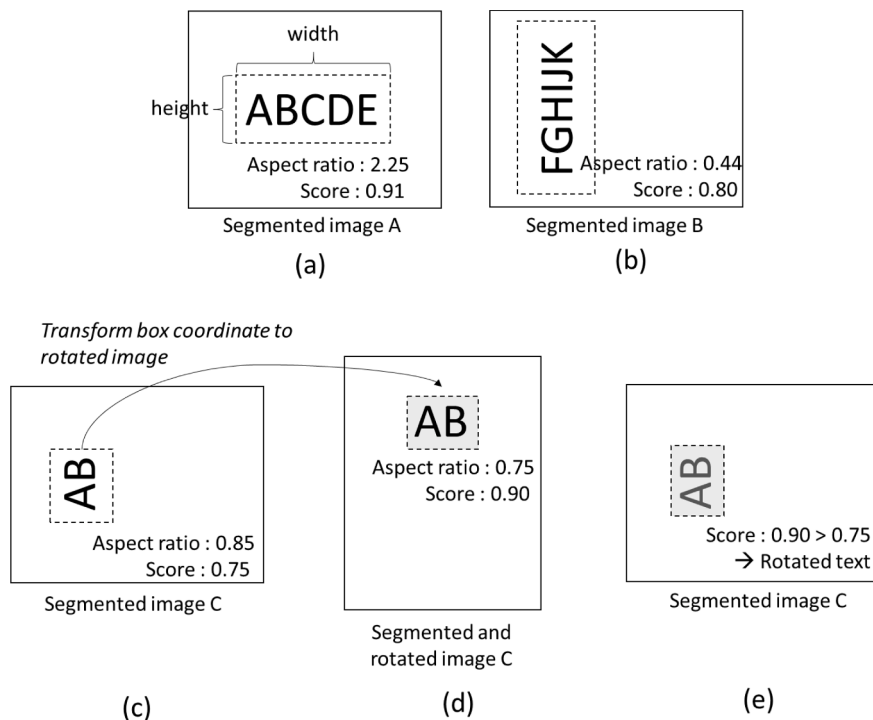


Fig. 11. Proposed rotation estimation method.

were then used as inputs to the rotation estimation process to determine whether each text was rotated. Finally, the recognition results for a character string in the image area within the text region using Tesseract, converting the recognition results according to the original P&ID coordinates, and merging the results.

The rotation estimation for a box was divided into two stages: rotation estimation based on the aspect ratio (i.e., width/height) and that based on the text recognition score. In aspect ratio-based rotation estimation, an aspect ratio of the text region of ≤ 0.5 indicated a rotated text string. In score-based rotation estimation, an aspect ratio of > 0.5 indicated a rotated text string if the rotated image had a higher score from the string recognition compared to the score from unrotated image.

For long text strings, a rotated text string can be easily determined by the aspect ratio, as in Fig. 11(a) and (b); however, for a short vertical string, it is difficult to determine the rotated text string only using the aspect ratio (Fig. 11(c)). In this case, when the text recognition scores in the rotated image are considered, as in Fig. 11(d), a higher score for the

rotated image indicates a rotated text string, and the recognition results from the rotated image is used as the final result for text recognition.

5. Saving and verifying symbol and text recognition results

The data structure for saving recognized symbol, text, and line objects was defined in extensible markup language (XML) format. Fig. 12 presents the data structure for the storing of recognized P&ID symbol and text objects. The `<symbol_object>` tag consists of seven sub-tags used to store recognition data for both symbol and text objects, and the `<id>` tag is used to write the recognition id for each object created sequentially in the P&ID recognition process. The `<type>` tag stores the details of the type of each object: `equipment_symbol`, `pipe_symbol`, `instrument_symbol`, and `text`. The `<class>` tag stores the labels of symbol objects and the content of text objects. The `<bndbox>` tag stores the *x* and *y* coordinates of the upper-left and lower-right points that constitute the recognized bounding box for each object. The `<degree>` tag stores the rotational information of a recognized object. The degree increases

```

<symbol_object>
  <id> </id>
  <type> </type>
  <class> </class>
  <bndbox>
    <xmin> </xmin>
    <ymin> </ymin>
    <xmax> </xmax>
    <ymax> </ymax>
  </bndbox>
  <degree> </degree>
  <flip> </flip>
  <etc> </etc>
</symbol_object>

```

Fig. 12. Data structure for recording the object recognition results from P&IDs.

to 90° from the horizontal axis via rotation in a counterclockwise direction. The <flip> tag is used when the symbol objects are flipped over from the normal position. The <etc> tag is dedicated to storing additional information depending on the engineering context.

This XML format was employed to record the symbol and text recognition results. Based on the class label from the symbol detection results, the dictionary saved in the class label was consulted in order to search for the symbol name and type, which led to the data results being recorded in the <type> and <class> tags. The bounding box of the symbol detection results was converted using the input P&ID coordinates from the combination process as a reference, so the resulting values were recorded in the <xmin>, <ymin>, <xmax>, and <ymax> tags. This study did not detect the presence or absence of <degree> or <flip> in the symbol recognition process, so these were not recorded. Future research can be conducted on the detection performance for the presence or absence of <degree> or <flip>. The text string data from text recognition results were recorded in the <class> tag and an attribute value “text” was recorded in <type> tag to indicate the result is text. The prediction of the rotated text string via the rotation estimation process was recorded as 90 in <degree> or 0 otherwise. Because <flip> was not part of the text data, a default value of 0 was recorded.

The XML files can be employed by users to verify the recognition

results and modify them if necessary. At the end of the P&ID recognition process, users need to verify the recognition results and correct them if there are any errors to enable further usage of P&ID information. An example of the modification tool is presented in 6.2.3, which is developed based on the XML format we defined. After verification and modification, XML file can be used as input for various engineering purposes, including P&ID digitalization and calculating the quantity of equipment, piping components, measuring devices, and pipelines.

6. Experiments

6.1. Experimental data

To secure sufficient dataset for deep learning, digital diagrams are required so that accurate symbol attributes and shapes can be obtained. Therefore, 82 digital P&IDs were modeled using the SmartPlant P&ID program based on 60 image-based P&IDs provided by a local Korean company. Symbol data in the P&IDs were extracted directly from SmartPlant P&ID using the in-house program developed in this study. As a result, a P&ID symbol dataset for 6,821 symbols was constructed based on three symbol categories: equipment, piping (piping components, nozzles, and OPCs), and instruments. A total of 76 symbol labels exist in the dataset. Table 6 presents an example of the elements of the generated P&ID symbol dataset. The P&ID symbol dataset was constructed to comply with the data structure used for symbol detection (Fig. 2a).

The image size of each symbol was randomly scaled after movement within an allowable range on the xy-plane for data augmentation to obtain sufficient training data. As a result, 10 additional images were obtained for each symbol, and a total of 75,031 datapoints for the symbols were generated.

For the text dataset, 10,073 images were extracted from the 82P&IDs. The P&ID text dataset complied with the data structure used for character detection (Fig. 2c). Table 7 presents representative text images generated from the P&IDs.

6.2. Experimental settings

In the P&ID symbol recognition experiment, learning was based on the bounding box coordinates of the symbols in 73 diagrams, excluding 9 test diagrams and the XML file listing the symbol labels(classes). The training, validation, and test datasets were split in the same manner as described for the preliminary experiment in Section 4.2.1.

The performance was measured for each experimental condition after 20 learning epochs for the small-symbol network and 150 epochs for the large-symbol network. The loss was confirmed to have converged sufficiently for this number of epochs. Learning was conducted in a

Table 6
Representative symbol images by symbol category.

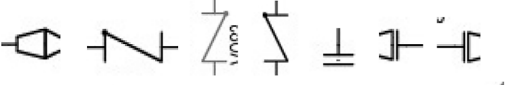
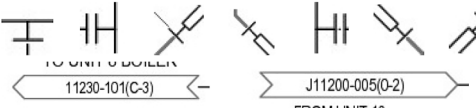
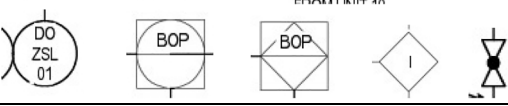
No	Type	Representative Symbol Images	Number
1	Equipment		319
2	Piping	Piping Component	2,204
	Nozzle		175
	OPC		481
3	Instrument		3,642

Table 7
Generated representative text images.

No	Diagram name	Representative text images	Number
1	A-22300-001-01	M-PP-01F, TO OIL SUMP, V024, V034, 082A-GBY-250, 018-GYP-100, 018-GYP-100, 086B-GBY-100	105
2	A-71120-001-02	TI, 11D, 150, XA, PT, LG, 04D, PSV, 06D,  , SEE VENDOR DWG NO. PID-00001-001, 971110-001(A-6), 200X150, 11	69
3	A-71120-001-03	10-001B-HBY-200, 10-V003, 10-V002, FC, 10-V001, AS, 10-XA, PV, 01, 971120-002(B-2), 001A-HBY-200, 916320-001(D-2)	225
4	B-36130-002-01	CD, PI, 10, AW, 205, DT, 02, BOP, (NOTE 3), BOP,  , 11A, 07B, TW, FT, BOP, PSL, TT	451
5	B-36130-002-02	M-HT-05, GLAND STEAM CONDENSER, CD, PT, J53700-002(D-5), 023J-HBD-25, J53700-002(D-3), J36130-001(D-4)	565

Table 8
Parameter list for P&ID symbol detection.

Parameters	Value	Note
Diagram resolution	9933 × 7016	
Segmentation resolution	800 × 800	Small-symbol network
Segmentation stride	300	Small-symbol network
Resize scale factor	0.125	Large-symbol network
Line thickening	Erosion (kernel size 2 × 2; 4 iterations)	Large-symbol network
Minimum score threshold	0.5	
Base augmentation	Horizontal flip (p = 0.5)	Small-symbol network
	Horizontal and vertical flip (p = 0.5)	Large-symbol network
$T_{overlap}$	0.2–0.5	Value depends on the symbol class
T_{else}	0.0	

Table 9
Parameter list for P&ID text recognition.

Parameters	Value
Diagram resolution	9933 × 7016
Segmentation resolution	2000 × 2000
Segmentation stride	1000 × 1000
Text region score threshold	0.7
Text region margin	(1 + 0.05) × Min (width, height)

Ubuntu environment using an Nvidia 1660Ti GPU. The Adam optimizer was used for the learning process, with learning rates of 0.00035 and 0.0001 for the small- and large-symbol networks, respectively. Other parameters for the experiment are listed in Table 8. The Adam optimizer was selected by comparing the performance results for an SGD optimizer with a learning rate of 0.003, a momentum of 0.9, and a weight decay of 0.0001. The use of the Adam optimizer increased the precision by 1.11% and the recall by 0.19% for the test dataset.

The parameters used in the text recognition experiment are listed in Table 9. The resolution of the original diagrams was identical to that for symbol detection. Recognition was conducted after dividing the entire diagram into 1000 strides with a resolution of 2000 × 2000. A score threshold of 0.7 was applied to the text region detection, Tesseract was used for each text region, and a margin weight of 0.05 was used for text recognition (i.e., text recognition was conducted in a region that was slightly larger than the detected text region).

6.3. Experimental results

6.3.1. Symbol detection results

The P&ID symbol detection experiment was conducted using the proposed symbol detection method. Quantitative performance assessment was based on 1) the precision and recall calculated from the derived detections with a score of 0.5 and an IOU threshold of 0.5 as references, 2) AP, AP₅₀, and AP₇₅, which are the average precision measures used in the COCO dataset, and 3) the average precision based on the diagonal length of the symbols: small (<150), medium (150–250), and large (>250).

In the experiment, we divided the proposed method into four steps to investigate its performance: 1) GFL (default), the performance when only the GFL-based small-symbol network was used; 2) GFL (A.T.), the performance when the adaptive threshold was applied to the small-symbol network results; 3) GFL (A.T.) + large, the performance when the large-symbol network was used in parallel; and 4) GFL (A.T.) + large w/augmentation, the performance when data augmentation was applied. These experiment steps were designed to quantitatively prove the effectiveness of each of our proposed procedures. By comparing GFL (default) with GFL (A.T.), we can confirm the effect of adaptive thresholding. By comparing GFL (A.T.) with GFL (A.T.) + large, we can confirm the effect of adopting a large-symbol network. GFL (A.T.) + large w/augmentation was tested to confirm the effect of additional improvement. In the last step, we experimented with cropping, upscaling, downscaling, and adding blur or noise, in addition to basic flip augmentation. Although adaptive thresholding in NMS may also be applied to large-symbol networks, the test P&IDs did not include self-overlapping large symbols. Thus GFL + large with and without adaptive thresholding produced identical performance results.

The performance results for the four steps of our proposed approach are presented in Table 10. The strongest performance for each measure is bolded and underlined, while the second-best performance is bolded and italicized.

The results in Table 10 show that using the large-symbol network, using an adaptive threshold, and augmentation all contributed to increasing the precision and recall performance. The complete GFL (A.T.) + large w/augmentation system had the highest precision and recall performance, with 0.9718 and 0.9827, respectively. In terms of AP performance, GFL (A.T.) + large exhibited a high overall performance. It should be noted that further hyperparameter tuning on the small- and large-symbol networks could improve the performance measures. Nevertheless, the focus of our experiment was to confirm the baseline usability of GFL (default) and to increase its performance with adaptive thresholding and the separation of the large-symbol network. Our experimental results produced a precision of 97% and a recall of 98% in

Table 10
Symbol detection performance.

	Precision	Recall	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
GFL (default)	0.9656	0.8911	0.8250	0.8610	0.8340	0.8740	0.8210	0.7980
GFL (A.T.)	0.9603	0.954	0.8330	0.8690	0.8420	0.8830	0.8210	0.8160
GFL (A.T.) + large	0.9604	0.9619	0.8550	0.8990	0.8720	0.8830	0.8210	0.8780
GFL (A.T.) + large w/augmentation	0.9718	0.9827	0.8350	0.9030	0.8880	0.8660	0.7910	0.8440

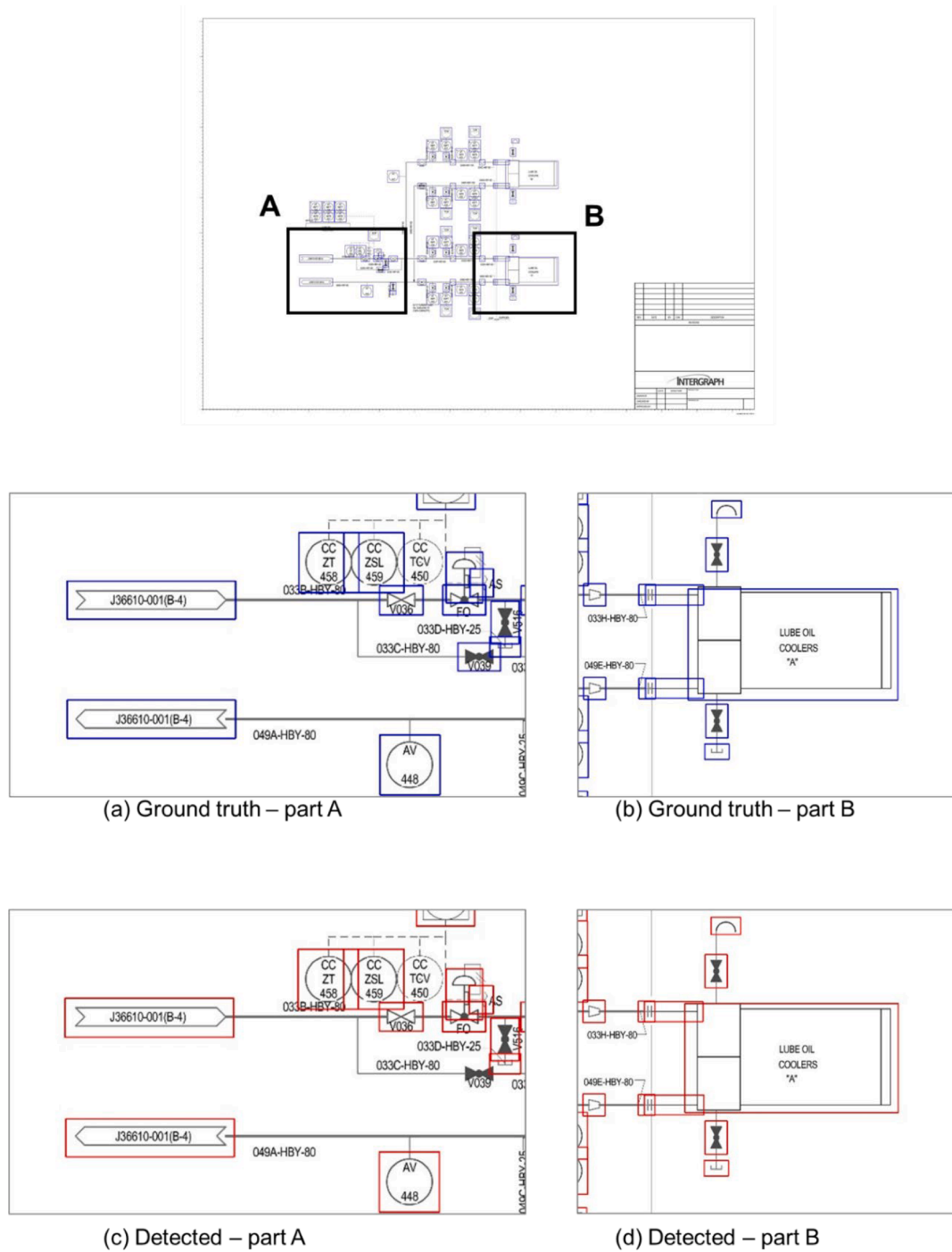
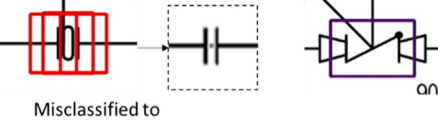
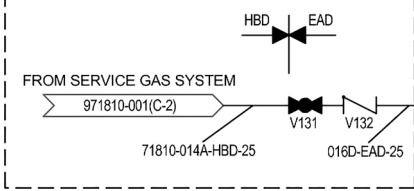
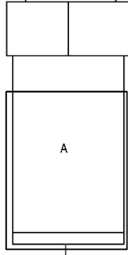


Fig. 13. Representative symbol detection results.

Table 11
Cases of failure in symbol detection.

Small symbols		Large symbols	
Misclassification		Not detected	Regression error
			
			

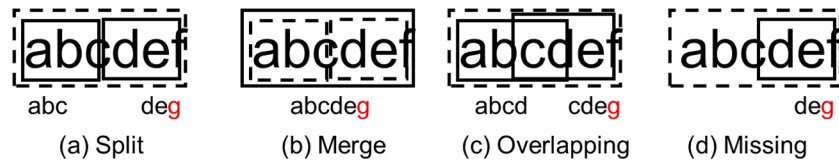


Fig. 14. Cases for penalization in CLEval.

Table 12
Text recognition performance.

Diagram	Detection		End-to-end	
	Precision	Recall	Precision	Recall
1	0.9985	0.9001	0.9652	0.9001
2	0.9422	0.9133	0.8549	0.9151
3	0.9845	0.9010	0.9230	0.9141
4	0.9903	0.9522	0.9561	0.9394
5	0.9242	0.9360	0.8711	0.8933
6	1.0000	0.9984	0.9901	0.9527
7	1.0000	0.9522	1.0000	0.9140
8	0.9917	0.9188	0.9485	0.9112
Average	0.9789	0.9340	0.9386	0.9175

Table 13
Detection and end-to-end attributes of the individual diagrams.

Diagram	Number of GT boxes	Detection Attributes			End-to-end Attributes	
		Split	Merge	False positives	Missing characters	Overlapping characters
1	117	0	0	2	74	0
2	107	0	0	21	47	0
3	97	0	1	5	91	0
4	156	1	1	5	52	0
5	55	0	0	36	50	0
6	165	1	0	0	30	0
7	49	0	0	0	27	0
8	161	0	0	3	70	0
Average	113.38	0.25	0.25	9.00	55.13	0.00

the experiment without further hyperparameter tuning.

The reason for the lower AP with augmentation could be because AP-related performance measures are calculated based on the mAP average for each class, so it may be challenging to obtain a consistent performance measure for a dataset with a large class imbalance such as the one in the present study. For example in our test diagram, only one or two instances were present for most symbols in a single diagram, though others had as many as 15 instances, such as flanges. Naturally, the detection error for symbols with one or two instances can affect the performance results more significantly in AP performance estimation. Secondly, training the network with augmented data decreases the

accuracy of bounding box regression while increasing detection precision and recall. With augmentation, we found that AP₉₀ and AP₉₅ performance was significantly lower compared to without augmentation. Therefore, overall AP (the average AP calculated with an IOU of 0.50 to 0.95) was lower with augmentation. In summary, it is suggested that, if accurate bounding box coordinates are required, data augmentation should be avoided. However, if there is a need to find as many instances as possible in a P&ID, augmentation increases the precision and recall performance.

The recognition results for a representative test diagram are presented in Fig. 13; Fig. 13(a) and (b) show the ground truth symbol data and the visualization data for each symbol's location in the diagram, respectively. Class data was also derived for each symbol, but they are not presented in the diagrams. Of the results estimated using the proposed symbol recognition network, Fig. 13(c) and (d) display the individual results for regions identical to those in Fig. 13(a) and (b). Most of the symbols and symbol classes were detected. The smallest symbol in the corresponding region had a height and width of 66 pixels, while the largest was 881 pixels. Using the diagram's horizontal resolution (9933) as a reference, symbols with sizes of 0.66%–8.87% were all detected. With the resolution of an image segment (800) as a reference, these symbols had sizes of 8.00% and 110.13%; the large symbol represented merged detection results from the large-symbol network.

Examples of failed cases are presented in Table 11. Misclassification occurred for small symbols, which were often mistaken for other symbols with similar shapes. Other failed cases were due to insufficient data in the training dataset. For large symbols, misdetection and a large regression error were the primary detection failures. The training dataset for the large-symbol network was too small, compromising the performance, though this was improved with augmentation. However, certain symbols did not reach a sufficient level of learning even after augmentation. Large symbols that were undetected using our proposed approach could become detectable if more data are used for training.

6.3.2. Text recognition results

A P&ID symbol recognition experiment was conducted using the proposed text recognition method. The performance measure proposed in CLEval (Baek et al., 2006) was applied to judge the performance of our approach. Unlike object detection, text recognition requires a combined performance estimation for text region detection and text recognition. A more comprehensive performance measure is also

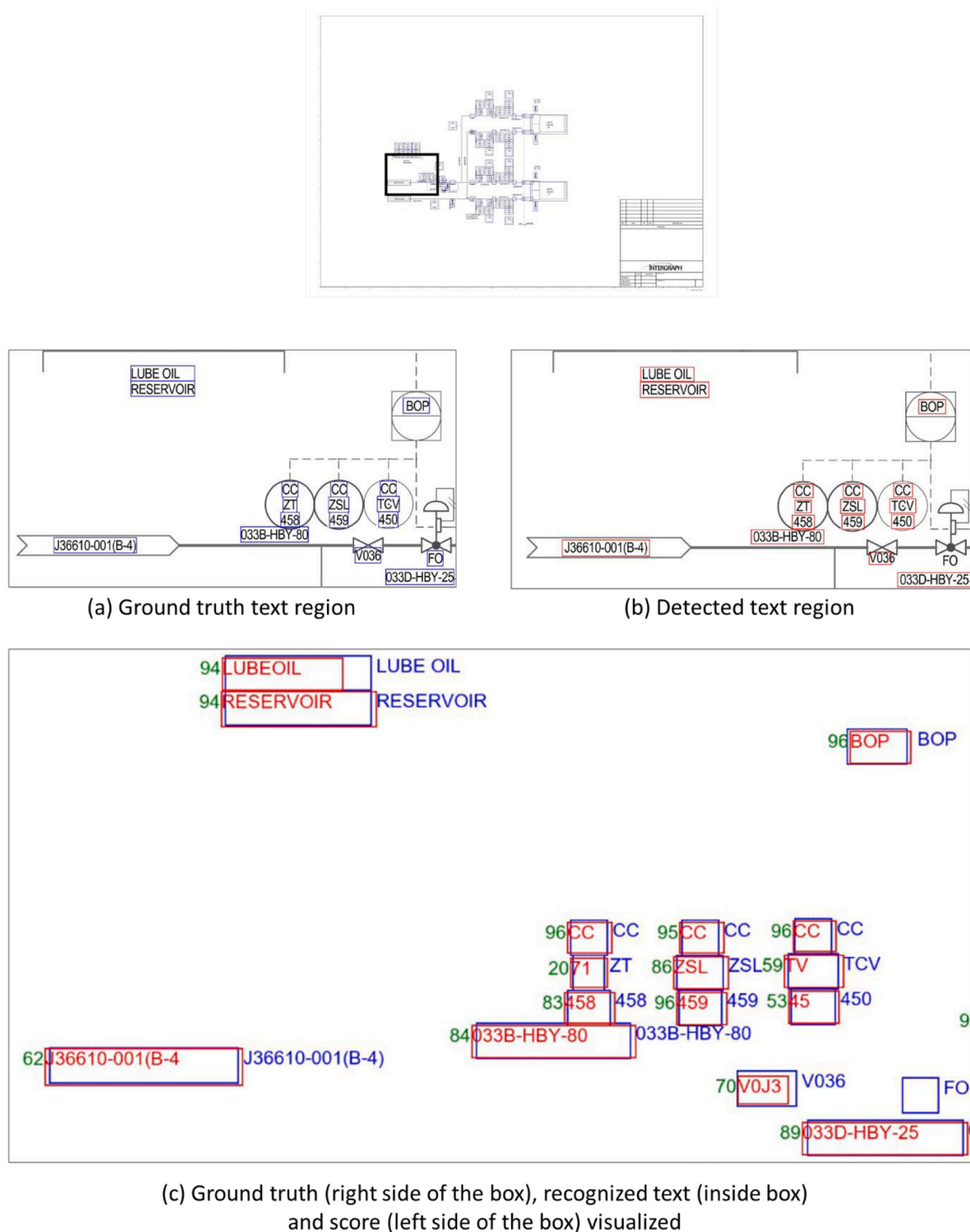


Fig. 15. Text recognition results.

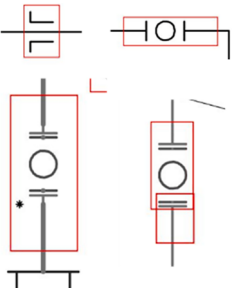
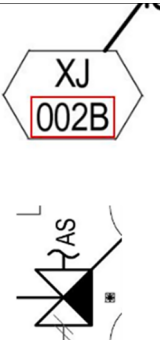
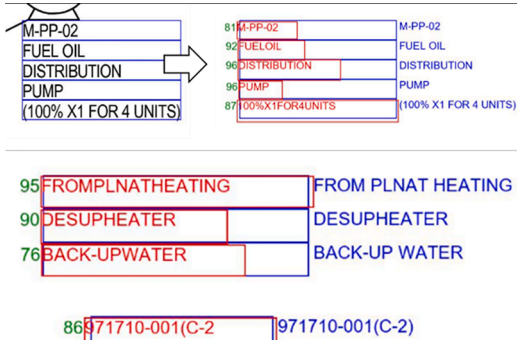
necessary because text is composed of multiple characters. Various performance estimation metrics have been proposed for text recognition to date. CLEval differs from previous approaches in that it divides performance estimation into matching and scoring to produce suitable precision and recall measures for various cases. In Fig. 14, the dotted lines and solid lines denote the ground truth and text region detection, respectively, with the recognition results labeled under the boxes. The performance for box detection and text recognition (end-to-end) was designed so that a penalty was incurred for cases such as splitting, merging, overlapping characters, or missing characters. For more detail

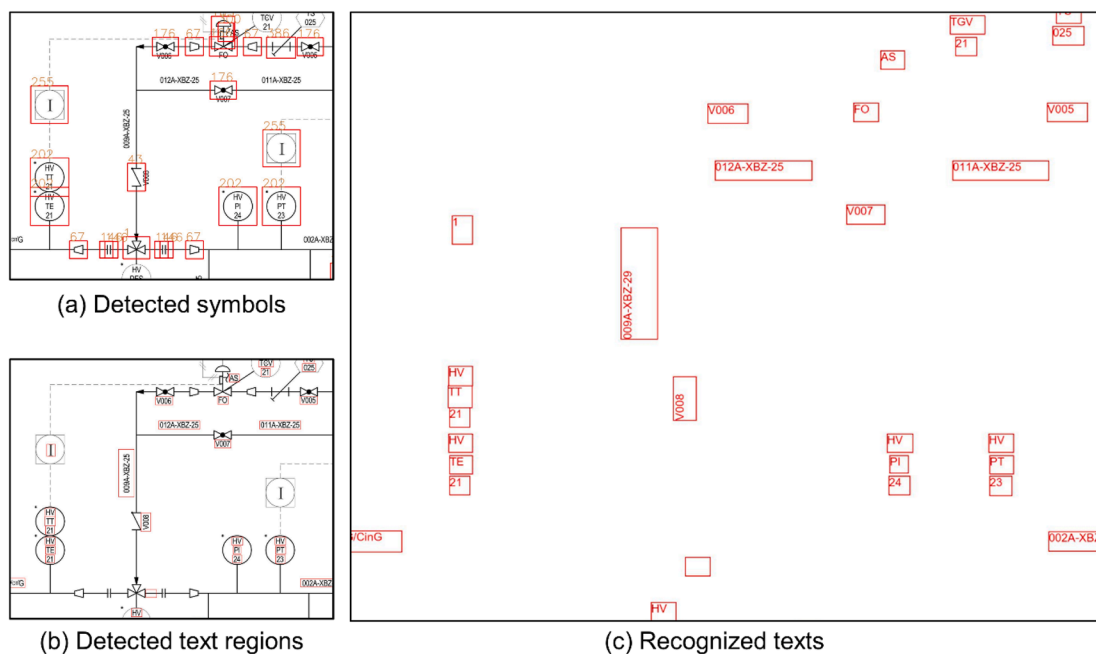
on the estimation method, see CLEval.

The text recognition performance for eight of the test diagrams—one was excluded because it did not have any text—was evaluated based on CLEval metrics as with symbol detection. An overview of the performance results is presented in Table 12. In the detection column, the average precision and recall for text region detection were 0.9789 and 0.9340, respectively. In the end-to-end column, the average precision and recall for text region detection and text recognition combined were 0.9386 and 0.9175, respectively. Each diagram showed large deviation in their performance result. Detailed attributes relating to the

Table 14

Cases of failure in the text recognition results.

False-positive text regions	Missing text regions	Ground truth box coordinate error
		

**Fig. 16.** Recognition results for a P&ID from a different source.

performance results are presented in Table 13. The number of false positives and missing characters were relatively high for test diagrams 5 and 3, respectively.

A visualization summarizing the key results from the test diagrams is presented in Fig. 15. In Fig. 15(a) and (b), the ground truth and detected text regions are shown, respectively. Fig. 15(c) visualizes the ground truth and detected results for one image with the symbols removed.

An analysis based on the visualization of the key factors contributing to lower text recognition performance was performed (Table 14). There were cases of false positives for the text region that misrecognized a symbol as a character, failure to recognize the text region, and errors in the ground truth box coordinates. Failure to recognize the text region primarily occurred with short text strings of less than 2 characters; missing characters thus occurred due to the failed recognition of the text region at the text detection stage. With errors in the ground truth box coordinates, the number of characters missing may increase even if the text is correctly recognized due to an incorrect box size. When the training data were generated, cases of ground truth boxes that were larger than the actual text string were observed. In CLEval's performance estimation, the pseudo-character center (PCC) calculation plays a significant role in matching and assumes equidistant characters. Thus,

the performance calculation may be incorrect if the ground truth box coordinates are inaccurate. Additionally, if a blank string is unrecognized during the text recognition process, all text strings after the blank are assumed to be missing, resulting in lower performance. We also confirmed an increase in the end-to-end recall to 0.9341 (3.5%) with the removal of the blanks in both the ground truth and the recognized text.

An additional experiment was performed for a P&ID separate from the established dataset (Fig. 16). The corresponding P&ID had more densely populated symbols than the training dataset. Considering the characteristics of the P&ID, the shape of the symbol and the text structure did not change significantly for diagrams drawn by the identical P&ID design program, even though the overall configuration of diagrams may be different. Thus, the symbol detection and the text recognition network trained using the training dataset established in our study was confirmed to be applicable to different P&IDs.

6.3.3. Data verification and modification results

The P&ID symbol and text recognition results were viewed and corrected using the prototype recognition result verification software. As presented in Fig. 17, the software provides a user interface for verifying and modifying the recognized objects. It consists of (a) a P&ID

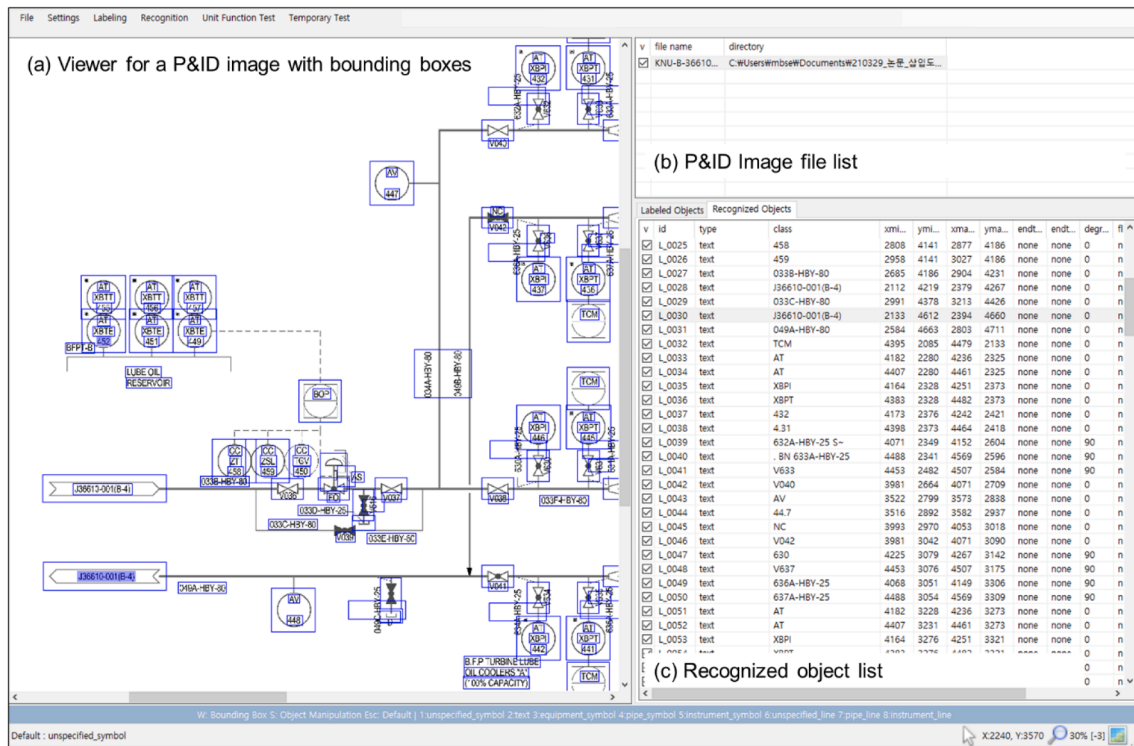


Fig. 17. Verification and modification of P&ID recognition results.

viewer at the left side, (b) a P&ID image file list in the top-right corner, and (c) a recognized object list in the lower-right corner. With the P&ID viewer, users can visually inspect the recognition results and modify or create bounding boxes. With the P&ID image file list, users can access the P&ID image file loaded onto the module. With the recognized object list, users can directly modify the attributes of a recognized object.

In the experiment, the output XML files and corresponding P&ID images were loaded onto the software module to display the recognition results. Any cases of failure found in the symbol and text recognition results were inspected and corrected using the P&ID viewer and the recognized object list. For each diagram, about 1800 s were required on average for data verification and modification.

6.3.4. Discussion

Symbol and text recognition methods were proposed based on the analysis of experimental results, and these demonstrated an excellent performance. It was shown that using a large-symbol network separation and an adaptive threshold improved symbol recognition performance. The AP performance measure was lower with augmentation due to the class imbalance, while precision and recall performance increased. As confirmed in our experiment, if the goal is to increase the number of correct instances and produce more accurate box regression results, using augmentation can improve the performance. We also confirmed that the GFL-based symbol detection method, proposed based on quantitative performance measures and visualized results, produced an excellent performance. However, the performance was relatively low for those symbols lacking training data and this requires improvement. Considering the characteristics of P&IDs, the larger the symbols, the scarcer the training data. Larger datasets and novel methodologies are required to resolve this issue and overcome the class imbalance in order to produce more robust results.

Existing text recognition methods exhibited a high performance. Errors in the ground truth box coordinates, which have a significant influence on lower performance, could be resolved simply by fixing errors in the program generating the training data. In contrast, the text and symbols with false positives or missing text regions consisted of lines and

curves, so feature extraction for the recognition process is challenging. This problem can be resolved by reinforcing negative sampling in learning. In the future, a combined network structure that can learn the relationship between the symbol recognition results and the text recognition results could improve overall text and symbol recognition performance.

7. Conclusion and future work

We proposed a method for recognizing symbols and texts in P&ID images using deep-learning technology. It consisted of pre-processing the P&ID image, symbol and text recognition, and storing the recognition results. In the pre-processing step, the outer borders and title box were removed to increase the accuracy of the object recognition process for P&IDs. The symbol recognition in P&IDs was based on the GFL method. Large- and small-symbol detection networks were implemented individually, but their results were combined, and adaptive NMS improved their performance. For text recognition in P&IDs, text region detection was conducted using the easyOCR framework, and then text recognition using Tesseract was employed. Additionally, the P&ID symbol taxonomy of 300P&ID symbols was defined using symbol classification information provided from the ISO 10628-2 and PIP PIC001. We also defined the structure of the deep neural network training data for P&ID recognition. To verify the proposed method, we experimented with symbol and text recognition using 82P&IDs remodeled based on data provided by a local Korean company. The resultant recognition performance was excellent for both symbols and texts. The precision and recall values were 0.9718 and 0.9827 for symbol recognition and 0.9386 and 0.9175 for text recognition, respectively.

Our study applied deep learning technology to P&IDs, which are characterized by their high density, with multiple symbols of various sizes and with complex shapes, to recognize the symbols and text at an industrially applicable level. Additionally, the training data structure and the standardization of symbol taxonomy were studied to improve the applicability of P&ID recognition technology.

Future research on the following topics is suggested. There is a lack

of P&ID training data for large symbols; a potential recognition method needs to be investigated to resolve this problem. Abundant training data not only improves the performance of large-symbol detection but also ameliorates performance degradation due to overfitting. Additionally, whether a combined network structure that can learn the relationships between the symbol recognition results and the text recognition results can improve recognition performance needs to be verified. Further, incremental learning methods should be investigated for the gradual improvement of deep neural networks based on training data obtained from numerous projects. Finally, we plan to collect more training data from scanned P&IDs and test the proposed method on scanned P&ID documents that contain noise and transformations.

CRedit authorship contribution statement

Hyunki Kim: Methodology, Software, Writing - original draft, Writing - review & editing. **Wonyong Lee:** Software. **Mijoo Kim:** Software. **Yoochan Moon:** Software. **Taekyong Lee:** Software, Writing - original draft. **Mincheol Cho:** Data curation, Writing - original draft. **Duhwan Mun:** Conceptualization, Methodology, Supervision, Writing - original draft, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eswa.2021.115337>.

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